

Perfect Circle — Business Testing Documentation

Prepared by: Manya Bansal · Sep 27, 2026 · Version 0.1

Real players run 33 acceptance scenarios on 7 device types, connected to the local server on the host PC; the game launches only when at least 95% pass, no Critical or High defects remain open, and Manya Bansal signs off.

Objectives

- Confirm every Must requirement works for real players on phones, tablets and laptops.
- Check that scores feel fair to players, and tune the strictness value k if they do not.
- Confirm the server starts on the host PC, and that phones and laptops on the same Wi-Fi can reach it, install it and play offline.

Team and environment

Role	Who	Responsibility
UAT lead and approver	Manya Bansal	Plans testing, triages defects, signs off launch
Testers	8 to 10 students	Run scenarios, fill the feedback form
Classroom tester	1 teacher	Runs the smartboard scenario (BT-29)
Developers	Project team	Fix defects, rebuild and restart the local server

Environment: the host PC runs the server with a separate test database file (`data/test.db`). All test devices join the same Wi-Fi and open `http://<PC IP address>:3000`. A test-only date override lets testers check streaks without waiting days.

Device matrix

Device	Browser	Input	Main focus
Android phone, mid-range	Chrome	Touch	Performance, install, share sheet

Device	Browser	Input	Main focus
iPhone, iOS 16 or later	Safari	Touch	Install from Share menu, scrolling not triggered while drawing
Tablet (iPad or Android)	Safari or Chrome	Two-finger touch	Split-screen duel
Windows laptop	Chrome and Edge	Mouse and trackpad	Turn-based split duel, keyboard use
MacBook	Safari	Trackpad	Safari differences, install
Smartboard or touch laptop	Chrome	Touch or pen	Classroom play
Host PC (Windows or macOS)	Chrome or Edge at localhost	Mouse	Server start, install at localhost, data kept after restart

Scenarios

ID	Scenario	Acceptance criteria	Requirement
BT-01	New player opens the link and draws a circle	Score shown within 1 s of finishing; no sign-up asked	FR-01, FR-03, FR-22
BT-02	Player draws each of the 4 shapes	Each is scored; ideal shape overlay matches the chosen shape	FR-01, FR-04
BT-03	Player draws too close, backwards, and not closed	Each shows its own clear message and a Retry button	FR-05
BT-04	Player plays time limit mode at 3 s	Countdown ring visible; attempt fails at 0	FR-06
BT-05	Player plays disappearing ink	Ink fades while drawing; full stroke shown on Result	FR-07
BT-06	Player plays moving dot	Dot moves smoothly; a steady circle around it scores well	FR-08
BT-07	Player turns on non-dominant hand	Result and history show the off-hand tag	FR-09

ID	Scenario	Acceptance criteria	Requirement
BT-08	Two phones: one creates a duel, the other joins by code	Both see each other's nickname within 3 s	FR-10, FR-11
BT-09	Second phone joins by scanning the QR	Joins the same room	FR-11
BT-10	Two players play a full best-of-3 duel	Same shape, synced countdown, same final result on both screens	FR-12
BT-11	One player turns off Wi-Fi mid-duel	Other player wins after 15 s with a clear message	FR-12
BT-12	Two players draw at once on a tablet in split screen	Both strokes drawn and scored separately	FR-13
BT-13	Two players take turns on a laptop with a mouse	Turns alternate; higher score wins	FR-14
BT-14	Two testers on different devices open the daily challenge	Same number, shape and mode	FR-15
BT-15	Player tries the daily challenge twice	Second try marked as practice, not counted	FR-16
BT-16	Player shares the daily result to WhatsApp	Text arrives with number, score and 5 squares, formatted correctly	FR-17
BT-17	Player completes dailies on 3 days (date override), then skips 1	Streak shows 3, then resets; best streak stays 3	FR-19
BT-18	Player scores 90% or more for the first time	Sharp eye badge pop-up appears once	FR-20
BT-19	Player opens Profile after 20 attempts	Graph shows attempts; filters by shape and mode work	FR-21
BT-20	Player logs in with nickname and PIN on a phone, plays, then logs in on a laptop	Same history, streak and badges on laptop	FR-23
BT-21	Install on Android (server on local HTTPS)	App icon on home screen; opens full screen	FR-26

ID	Scenario	Acceptance criteria	Requirement
BT-22	Install on iPhone via Share, Add to Home Screen (server on local HTTPS)	Opens full screen with the app icon	FR-26
BT-23	Install on the host PC from the browser's install button at localhost	Opens in its own window	FR-26
BT-24	Load the app, turn on airplane mode, play classic	Solo play and history work offline; syncs when back online	FR-27
BT-25	Draw quickly on a mid-range phone	No visible lag or broken line	NFR-01
BT-26	Each tester draws 10 shapes and rates score fairness 1 to 5	Average rating 4 or more	FR-03
BT-27	Switch the phone to dark mode	All screens readable in dark theme	FR-28
BT-28	Use menus with keyboard only on a laptop	Every button reachable with Tab and Enter	NFR-05
BT-29	Teacher runs split-screen duels on a smartboard with a class	Works on the large screen; students understand the score	FR-13
BT-30	Tester looks at a duel and a leaderboard	Only nicknames shown; no PIN visible	NFR-07
BT-31	Start the server with one command on the host PC	Game opens at <code>http://localhost:3000</code> within 5 s	FR-29
BT-32	A phone on the same Wi-Fi opens the PC's IP address	Game loads; login and duels work	FR-30
BT-33	Restart the PC, then start the server again	All players, history and badges are still there	NFR-08

Feedback form

Each tester answers on a scale of 1 to 5 after the session: the game is easy to start; the score feels fair; duels are fun; I would play the daily challenge again tomorrow; I would share my result. One open question asks what they would change.

Defect severity

Severity	Meaning	Example
Critical	Game unusable or data lost	App does not load on iPhone
High	A Must feature fails	Duel results differ on the two phones
Medium	Works, but wrong or confusing	Wrong message for a backwards stroke
Low	Cosmetic	Badge icon slightly misaligned

Exit criteria

- At least 32 of 33 scenarios pass.
- No open Critical or High defects.
- Average fairness rating (BT-26) is 4 or more.
- All 48 unit tests pass on the release build.

Sign-off

- ☐ Business testing complete and exit criteria met. Approved by Manya Bansal, product owner and UAT lead. Date and signature: _____